Exercise 6

Simulate the lifecycle stages for UI design using the RAD model and develop a small interactive interface.

AIM:

The aim is to demonstrate the lifecycle stages of UI design via the RAD model and develop a small interactive interface for a Smart Home System employing Axure RP.

PROCEDURE:

Tool Link: https://www.axure.com/, OpenProj

Simulating the Lifecycle Stages for UI Design Using the RAD Model The RAD model emphasizes quick development and iteration through the following phases:

1. **Requirements Planning:** Gather initial requirements, identify key features of the Smart Home UI, and engage stakeholders to define control parameters (e.g., lighting, temperature).
2. **User Design:** Create initial prototypes and wireframes for the control dashboard. Conduct feedback sessions and use Axure RP to build interactivity.
3. **Construction:** Develop the actual interface based on refined designs and perform iterative testing cycles.
4. **Cutover:** Deploy the final interface and provide user training for device synchronization.

Axure RP Interactive Interface Development

Phase 1: Requirements Planning

1. **Identify Key Features:**
    - Navigation (Dashboard, Room Selection, Device Controls, Energy Reports, Security Log, Settings).
    - User actions (Toggling lights, Adjusting Thermostat, Scheduling, Monitoring Cameras).
2. **Create a Requirements Document:** List all functionalities and document user stories (e.g., "As a user, I want to dim the lights via my phone").

Phase 2: User Design

1. **Install and Launch Axure RP:** Download from the official website and launch.

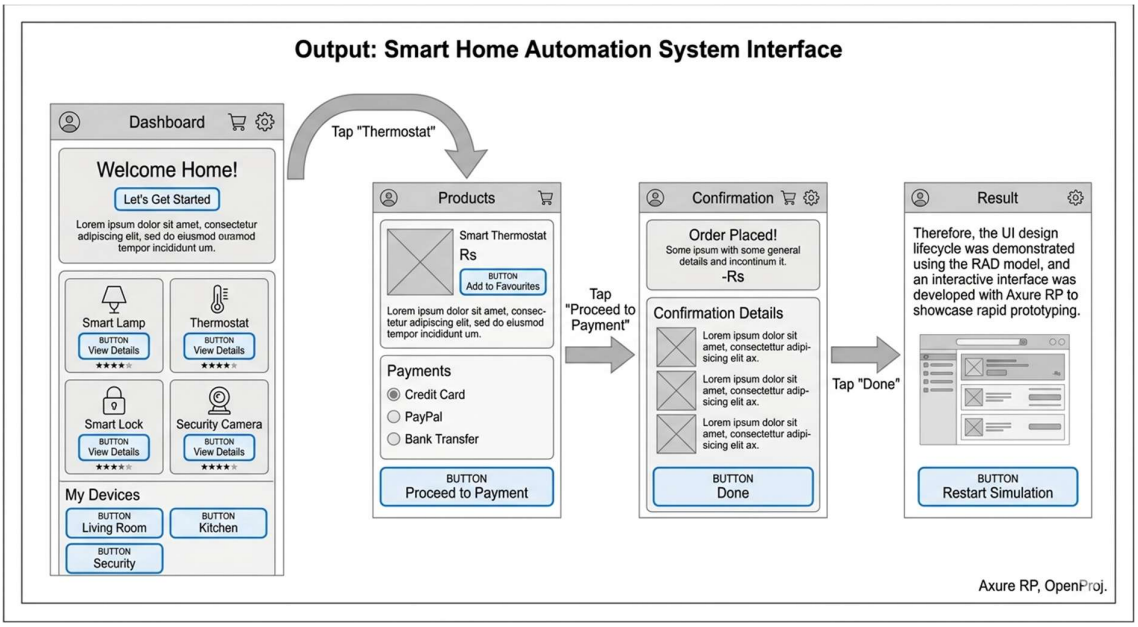
2. **Create a New Project:** Go to File -> New and name it "Smart Home Control Interface".
3. **Create Wireframes:** Drag widgets to design screens for:
    - Main Dashboard
    - Room View (Living Room, Kitchen)
    - Device Sliders (Brightness/Temp)
    - Security Status
4. **Add Interactions:** Select buttons (e.g., "Power On") and set an "OnClick" interaction to trigger state changes.
5. **Create Masters:** Build reusable headers and navigation bars to drag onto all wireframes.
6. **Add Annotations:** Use the Notes panel to describe sensor triggers and automated schedules.

Phase 3: Construction

1. **Develop Interactive Prototypes:** Use dynamic panels to create sliding menus and pop-up alerts for security breaches.
2. **Test and Iterate:** Preview the prototype, gather feedback on the "Ease of Use" for the thermostat slider, and adjust.

Phase 4: Cutover

1. **Finalize and Export:** Finalize transitions and export as an HTML file or share via Axure Cloud.
2. **User Training:** Conduct sessions to help users understand icon meanings and scheduling logic.



Result

Therefore, the UI design lifecycle was demonstrated using the RAD model, and an interactive interface for a Smart Home System was developed with Axure RP to showcase rapid prototyping.